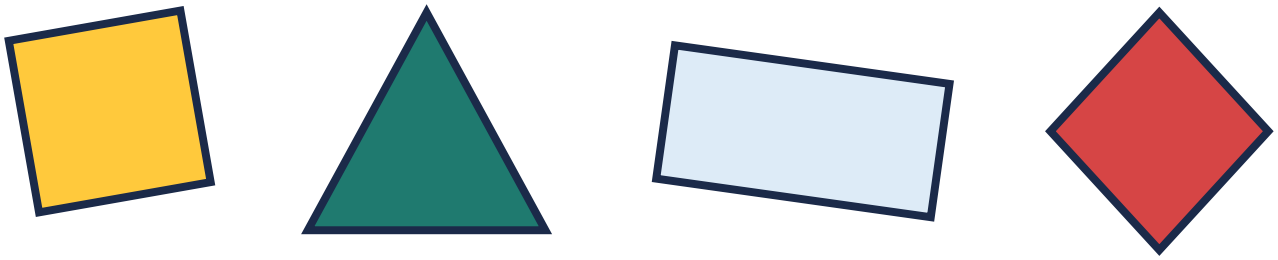
FOR PARENTS • PRIMARY 1

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, buttons, spoons, pasta or blocks. Try it with objects before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

The three levels

Core is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

What your child is doing in class

Ten 90-minute lessons over five weeks. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 LESSON 1 • NUMBER PATTERNS
Count with the Frog

Core

1 Write the next two numbers and the rule: 4, 7, 10, 13, __, __

2 Write the next two numbers and the rule: 5, 10, 15, 20, __, __

3 Write the next two numbers and the rule: 1, 5, 9, 13, __, __

4 Write the next two numbers and the rule: 50, 45, 40, 35, __, __

Think Harder

5 Fill in the missing numbers: 3, 8, __, 18, __, 28

6 Fill in the missing numbers: 40, __, 28, __, 16, 10

7 A pattern starts at 3 and adds 5 each time. What is the 8th number?

What your child learns

Counting in 2s, 3s and 5s, and finding the rule of a pattern that goes up or down.

Tip for you

Hop a coin along a paper number line. Ask your child to say the rule out loud before writing the next number.

Ask

"What is the rule? Does it work for every pair?"

Challenge

8 Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)

9 I am in the counting patterns of 3s and of 5s. I am more than 10 and less than 40. Which numbers can I be?

W1 LESSON 2 • NUMBER BONDS
Number Bond Inquiry

Core

1 10 = 3 + __

2 10 = __ + 8

3 10 = 6 + __

4 20 = 5 + __

Think Harder

5 5 + 4 = 6 + __

6 8 + 2 = 7 + __

7 9 + 6 = __ + 10

What your child learns

Number bonds of 10 and 20, and reading = as "is the same as" on both sides.

Tip for you

Split 10 buttons into two groups in every possible way. List them in order so none is missed.

Ask

"How do you know both sides are the same?"

Challenge

8 Put 1, 2, 3, 4, 5 and 6 in the triangle so each side adds to 9. Corners 1, 2 and 3. What goes in the middle of each side?

9 Corners 4, 5 and 6, and the numbers 1, 2, 3 in the middles. What do all the sides add to? Where does each middle number go?

W2 LESSON 3 • PICTURE GRAPHS
Let's Plan the Fruit Party

Core

1 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the most votes?

2 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the fewest votes?

3 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many more children chose orange than grape?

4 A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many children voted altogether?

Think Harder

5 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?

6 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?

What your child learns

Reading picture graphs where one picture can stand for 1, 2 or 3 children, comparing rows and finding totals.

Tip for you

Hold a family vote (favourite fruit or snack) and draw it together, one picture per vote.

Ask

"What does one picture stand for?"

7 1 picture = 2 children. Red has 3 pictures and blue has 5. How many more children chose blue than red?

Challenge

8 18 children voted. 7 chose apple and 5 chose orange. The rest chose grapes. How many chose grapes?

9 Votes: A 8, B 5, C 6. Then 4 more children join and all choose B. Which fruit wins now, and by how many votes?

W2 Pattern Inquiry

LESSON 4 • REPEATING AND GROWING PATTERNS

Core

1 The pattern is circle, square, triangle, circle, square, triangle, circle ... What is the 10th one?

2 The pattern is star, heart, star, heart, star, heart ... What is the 9th one?

3 The pattern is red, red, blue, red, red, blue, red ... What is the 8th one?

4 The pattern is A, B, C, D, A, B, C ... What is the 11th one?

Think Harder

5 Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square ... What is the 14th shape?

6 Same pattern: how many triangles are in the first 20 shapes?

7 Same pattern: in which places do the squares come? Write the first five.

What your child learns

Finding the part of a pattern that repeats, predicting far-away places, and growing patterns with a table.

Tip for you

Make a pattern with spoons and forks. Ask what the 10th one will be before you build it.

Ask

"Which part repeats? How many things are in it?"

Challenge

8 Squares in a row use 4 sticks for 1 square, 7 sticks for 2 squares and 10 sticks for 3 squares. How many sticks for 8 squares?

9 Using the same pattern, how many squares can you make with 22 sticks?

W3 LESSON 5 • WORKING BACKWARDS
Mystery Number

Core

1 I think of a number. I add 3. I get 8. What is my number?

2 I think of a number. I add 6. I get 14. What is my number?

3 I think of a number. I add 5. I get 12. What is my number?

4 I think of a number. I add 9. I get 17. What is my number?

Think Harder

5 I think of a number. I add 5, then I take away 3. I get 9. What is my number?

6 I think of a number. I add 6, then I take away 2. I get 10. What is my number?

7 I think of a number. I add 4, then I take away 7. I get 8. What is my number?

What your child learns

Working backwards: undoing each step, from the last step to the first.

Tip for you

Hide some counters under a cup, add a few more on top, and let your child work out how many were hidden.

Ask

"How can you check by going forwards?"

Challenge

8 I think of a number. I add 6, take away 4, then add 5. I get 17. What is my number?

9 A number machine has two steps: add 3, then add 4. IN 5 gives OUT 12. What is IN when OUT is 20?

W3 LESSON 6 • MONEY Coin Combos

Core

1 Pay exactly 15¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

2 Pay exactly 25¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

3 Pay exactly $15 with the fewest notes. The notes are $2, $5, $10 and $50.

4 Pay exactly $26 with the fewest notes and coins. You have $1 coins and $2, $5, $10 and $50 notes.

Think Harder

5 How many ways can you make 25¢ using 20¢, 10¢ and 5¢ coins?

6 How many ways can you make 30¢ using 20¢, 10¢ and 5¢ coins?

What your child learns

Finding every way to make an amount, and paying with the fewest coins and notes, in cents and in dollars.

Tip for you

Lay real coins and notes on the table. Ask "Can you do it with fewer?" before checking the answer.

Ask

"How do you know you found them all?"

7 How many ways can you make 20¢ using only 10¢ and 5¢ coins?

Challenge

8 Make 35¢ using exactly 3 coins from 5¢, 10¢, 20¢ and 50¢. How many ways are there?

9 Make 40¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find them all.

W4 LESSON 7 • 2D SHAPES
Shape Safari

Core

1 How many sides and corners does a triangle have?

2 How many sides and corners does a square have?

3 How many sides and corners does a pentagon have?

4 How many sides and corners does a hexagon have?

Think Harder

5 I have 4 sides. All my sides are the same length. What shape am I?

6 How many sides do 3 triangles have altogether?

7 How many corners do 5 squares have altogether?

What your child learns

Describing shapes by sides and corners, counting every shape in a picture, and making new shapes.

Tip for you

Fold and cut a paper square. What shapes can your child make by putting the pieces together again?

Ask

"Did you count the bigger shapes too?"

Challenge

8 A square has both diagonals drawn. How many triangles can you find?

9 A big triangle has 1 line from the top corner to the bottom. How many triangles? What about 2 lines? 3 lines?

W4 LESSON 8 • LOGIC AND SORTING
Logic Lab

Core

1 Ali is taller than Ben. Ben is taller than Mei. Who is the tallest? Who is the shortest?

2 The cat is heavier than the hen. The hen is heavier than the mouse. Who is the lightest?

3 Red, blue and green ribbons. Red is longer than blue. Green is shorter than blue. Which ribbon is the longest?

4 Which number is the odd one out: 2, 4, 6, 9, 10? Why?

Think Harder

5 In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?

6 Ana, Bo and Cy each have a different toy: a ball, a kite or a yo-yo. Bo does not have the kite. Cy has the ball. Who has which toy?

What your child learns

Using clues to put things in order, sorting with two rules, and solving small logic grids.

Tip for you

Play "Guess my number" from 1 to 20 with yes or no questions. Which questions help most?

Ask

"Which clue tells you something for sure?"

7 Numbers 1 to 12. Which do we say when counting in 2s? Which when counting in 3s? Which in both?

Challenge

8 Four friends stand in a line. Ali is not first. Ben is ahead of Ali. Sam is last. Mei is behind Ali. In what order do they stand?

9 Three boxes are labelled “Apples”, “Oranges” and “Mixed”. Every label is wrong. You may take one fruit from one box. Which box should you pick from, and why?

W5 LESSON 9 • LENGTH
Measure It!

Core

1 A pencil is 7 cubes long. A pen is 10 cubes long. How many cubes longer is the pen?

2 A ribbon is 12 cubes long. You cut off 5 cubes. How long is the piece left?

3 Ali's book is 9 cubes long and Mei's book is 6 cubes long. Whose is longer? By how much?

4 Which is longer: 8 cubes or 8 paper clips? Why can we not tell?

Think Harder

5 1 clip is as long as 3 cubes. A pen is 12 cubes long. How many clips long is the pen?

6 1 clip is as long as 3 cubes. A ruler is 5 clips long. How many cubes long is the ruler?

What your child learns

Measuring with equal units and a centimetre ruler, comparing lengths, and changing from one unit to another.

Tip for you

Measure a spoon with paper clips, then with erasers. Talk about why the two numbers are different.

Ask

"Why does a bigger unit give a smaller number?"

7 Ali's desk is 6 spans long. Mei's desk is 9 spans long, but they measured the same desk. Whose hand is bigger?

Challenge

8 A rope is 20 cubes long. Ali cuts it into pieces that are each 4 cubes long. How many pieces does he get? How many cuts does he make?

9 A train has 3 carriages. Each is 6 cubes long. There is a gap of 1 cube between carriages. How long is the whole train?

W5 Challenge Day

LESSON 10 • MIXED PUZZLES

Core

1 Count in 3s: 3, 6, 9, __, __

2 10 = 4 + __ and 20 = 12 + __

3 Red 4, blue 6, green 3. How many more are blue than green?

4 I think of a number. I add 4. I get 9. What is it?

Think Harder

5 A pattern starts at 5 and adds 3 each time. What is the 10th number?

6 Ali has 12 stickers. He gives 5 to Mei and then gets 3 more. How many does he have now?

7 Which three coins from 5¢, 10¢, 20¢ and 50¢ make 45¢?

What your child learns

Choosing how to start a puzzle: draw it, list it in order, or make a table.

Tip for you

When your child is stuck, ask "Can you draw it?" before giving any other hint.

Ask

"How did you start? What would you try next time?"

Challenge

8 Two numbers add to 12 and differ by 2. What are they?

9 1 triangle uses 3 sticks, 2 triangles use 5 sticks. How many sticks for 9 triangles? How many triangles with 25 sticks?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Count with the Frog

1. 16, 19. Add 3.
2. 25, 30. Add 5.
3. 17, 21. Add 4.
4. 30, 25. Take away 5.
5. 13 and 23. Add 5.
6. 34 and 22. Take away 6.
7. 38
8. 12, 24 and 36.
9. 15 and 30.

Week 1 • Lesson 2: Number Bond Inquiry

1. 7
2. 2
3. 4
4. 15
5. 3
6. 3
7. 5
8. Between 1 and 2: 6. Between 2 and 3: 4. Between 3 and 1: 5.
9. Each side adds to 12. Between 4 and 5: 3. Between 5 and 6: 1. Between 6 and 4: 2.

Week 2 • Lesson 3: Let's Plan the Fruit Party

1. Orange
2. Grape
3. 5 (9 – 4)
4. 25 (7 + 5 + 9 + 4)
5. 10
6. 24 (12 pictures, each worth 2: count in 2s)
7. 4 (2 more pictures, each worth 2: 2, 4)
8. 6
9. B wins with 9, which is 1 more than A.

Week 2 • Lesson 4: Pattern Inquiry

1. circle
2. star
3. red
4. C
5. triangle
6. 10. The core circle, triangle, triangle, square has 2 triangles. Count the cores in 4s: 4, 8, 12, 16, 20, so there are 5 cores. Count the triangles in 2s: 2, 4, 6, 8, 10.
7. 4, 8, 12, 16, 20
8. 25. Keep adding 3: 4, 7, 10, 13, 16, 19, 22, 25.
9. 7. Keep the table going: 4, 7, 10, 13, 16, 19, 22 sticks. 22 sticks make 7 squares.

Week 3 • Lesson 5: Mystery Number

1. 5
2. 8
3. 7
4. 8
5. 7
6. 6
7. 11
8. 10 (17 – 5 + 4 – 6)
9. 13 (the same as adding 7)

Week 3 • Lesson 6: Coin Combos

1. 5¢ + 10¢
2. 5¢ + 20¢
3. $10 + $5 (2 notes)
4. $10 + $10 + $5 + $1 (4 pieces)
5. 4 ways. (20+5; 10+10+5; 10+5+5+5; 5+5+5+5+5)
6. 6 ways.
7. 3 ways. (10+10; 10+5+5; 5+5+5+5)
8. 1 way: 20¢ + 10¢ + 5¢
9. 2 ways: 20¢ + 10¢ + 5¢ + 5¢; 10¢ + 10¢ + 10¢ + 10¢

Week 4 • Lesson 7: Shape Safari

1. 3 sides and 3 corners
2. 4 sides and 4 corners
3. 5 sides and 5 corners
4. 6 sides and 6 corners
5. A square
6. 9
7. 20
8. 8 (4 small and 4 made of two small ones)
9. 3, 6 and 10 triangles.

Week 4 • Lesson 8: Logic Lab

1. Ali is tallest; Mei is shortest.
2. The mouse
3. Red
4. 9. It is the only odd number.
5. Tia (order: Mei, Sam, Tia, Raj).
6. Cy: ball. Bo: yo-yo. Ana: kite.
7. Counting in 2s: 2, 4, 6, 8, 10, 12. Counting in 3s: 3, 6, 9, 12. Both: 6 and 12.
8. Ben, Ali, Mei, Sam.
9. Pick from the box labelled “Mixed”. Its label is wrong, so it holds only one kind of fruit. If you take out an apple, that box is Apples. The box labelled “Oranges” cannot be oranges, and apples are taken, so it is Mixed. The box labelled “Apples” must be Oranges.

Week 5 • Lesson 9: Measure It!

1. 3
2. 7 cubes
3. Ali’s, by 3 cubes.
4. We cannot tell. The units are different sizes.
5. 4 clips
6. 15 cubes
7. Ali’s. A bigger span needs fewer spans to cover the same desk.
8. 5 pieces and 4 cuts.
9. 20 cubes (18 + 2)

Week 5 • Lesson 10: Challenge Day

1. 12, 15
2. 6 and 8
3. 3
4. 5
5. 32
6. 10
7. 20¢ + 20¢ + 5¢
8. 5 and 7
9. 19 sticks for 9; 12 triangles for 25 sticks.